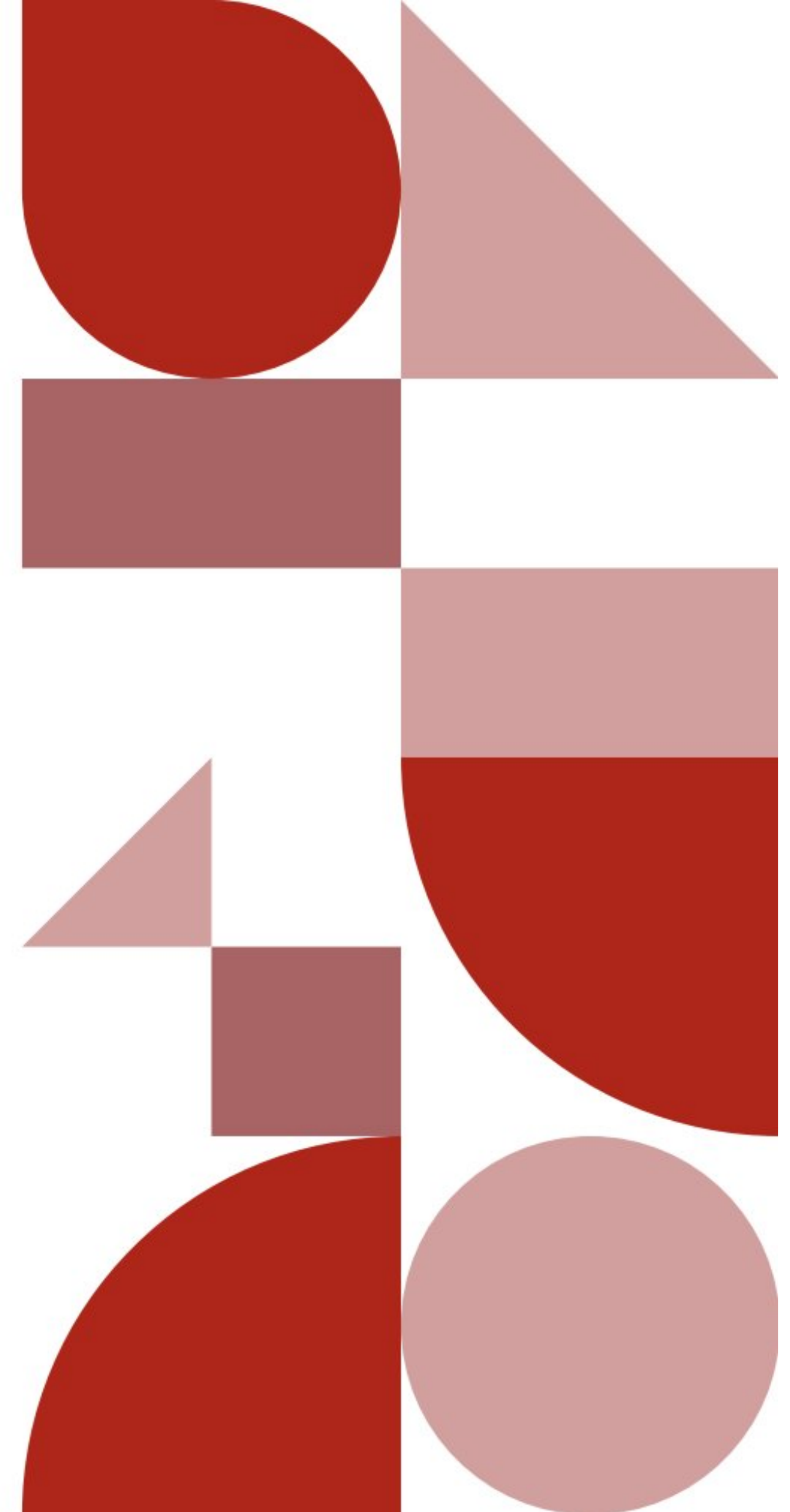




WEEK 8

CS50 INTRODUCTION INTO COMPUTER SCIENCE



Agenda

1 JavaScript Writing Methods

2 Input & Output Handling

3 JavaScript Basics

4 Objects in JavaScript

5 Arrays & Strings

6 The DOM (Document Object Model)

7 Event Handling

8 Edx Problems

JAVASCRIPT

JavaScript Writing Methods

JAVASCRIPT

1- Inline Example

```
<> WritingMethods.html > html
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4  |   <title>Inline JavaScript Example</title>
5  </head>
6  <body>
7  |   <button onclick="alert('Hello')">Click Me</button>
8  </body>
9  </html>
10
```

2- Internal Example

```
11
12  <!DOCTYPE html>
13  <html lang="en">
14  <head>
15  |   <title>Internal JavaScript Example</title>
16  |   <script>
17  |       function greet() {
18  |           alert("Hello, World!");
19  |       }
20  |   </script>
21 </head>
22 <body>
23 |   <button onclick="greet()">Click</button>
24 </body>
25 </html>
26
27
```

JAVASCRIPT

3- External Example

```
27
28 <!DOCTYPE html>
29 <html lang="en">
30 <head>
31   <title>External JavaScript Example</title>
32   <script src="script.js"></script>
33 </head>
34 <body>
35   <button onclick="greet()">Click Here</button>
36 </body>
37 </html>
38
```

```
JS script.js > greet
1  function greet() {
2    alert("Hello, World!");
3  }
4
```

JAVASCRIPT

Input & Output Handling

JAVASCRIPT

- Input & Output Example

JS Input.js > ...

```
1 let name = prompt("What is your name?");  
2 alert("Hello, " + name + "!");
```

INPUT

```
3  
4  
5 alert("This is an alert box!");
```

OUTPUT

```
6  
7  
8 console.log("This message is logged in");  
9  
10  
11 document.getElementById("p").innerHTML = "This is JS !";
```

JAVASCRIPT

JavaScript Basics

JAVASCRIPT

1- Variables

```
JS script.js > ...
1  variable
2  Using var
3
4  var name = "John";
5  console.log(name); // Output: John
6
7  Using let
8
9  let age = 25;
10 console.log(age); // Output: 25
11
12 Using const
13
14 const pi = 3.14;
15 console.log(pi); // Output: 3.14
16
17
```

2- For loop

```
19  for loop
20
21
22  for(var i =0; i<3; i++) {
23      console.log(i);
24  }
25
```

3- While loop

```
27  while
28  let j = 0;
29  while (j < 3) {
30      console.log(j); // Output: 0, 1, 2
31      j++;
32  }
33
34
```

JAVASCRIPT

4- Do-While loop

```
35  do-while
36  do {
37      console.log(i); // Output: 0
38      i++;
39  } while (i < 1);
40
```

5- IF Condition

```
42  if statement
43
44  let age = 17;
45  if (age >= 18) {
46      console.log("You are an adult!"); // Output: You are an adult!
47  }
48
```

6- IF-Else

```
50  //if-else
51  let age = 16;
52  if (age >= 18) {
53      console.log("You are an adult!");
54  } else {
55      console.log("You are a minor."); // Output: You are a minor.
56  }
```

7- IF-Else if

```
59  else if
60  let score = 70;
61  if (score >= 90) {
62      console.log("Grade: A");
63  } else if (score >= 80) {
64      console.log("Grade: B"); // Output: Grade: B
65  } else {
66      console.log("Grade: C");
67  }
```


JAVASCRIPT

8- Switch

```
72 //switch
73 let grade = 'H';
74
75 switch (grade) {
76     case 'A':
77         console.log("Excellent");
78         break;
79     case 'B':
80         console.log("Good");
81         break;
82     case 'C':
83         console.log("Fair");
84         break;
85     case 'D':
86         console.log("Poor");
87         break;
88     case 'F':
89         console.log("Fail");
90         break;
91     default:
92         console.log("Invalid grade");
93 }
94 // Output: Good
```

9- Functions

```
98 //Function Declaration
99
100 function greet(name) {
101     console.log("Hello" + " " + name)
102 }
103
104 console.log(greet("Alice"));
105
106
107
108
109 //Function Expression
110
111 const add = function (a, b) {
112     return a + b;
113 };
114 console.log(add(5, 3)); // Output: 8
115
116 💡
117
118 //Arrow Functions
119
120 const multiply = (x, y) => x * y;
121 console.log(multiply(4, 2)); // Output: 8
```